HOUR: 1

1. Mercury atom has 80 protons and 121 neutrons. Write the isotopic notation for mercury.
MID SEM 2012 / 2013 (CLO1, C1)

2. Complete the following table. **(CLO1, C1)**

Element	No of subatomic particles			Nucleon number	Isotopic notation
	p	e	n		
Silicon		14		28	
Chlorine					$^{35}_{17}Cl$
Bromine	35	36	44		
Magnesium	12	10		24	

3. An ion **Y^{3+}** has 18 electrons and 24 neutrons.
   a) Determine the proton number and nucleon number of the element **Y**.
   b) Write the isotopic notation to represent the element. **MID SEM 2015 / 2016 (CLO1, C1)**

4. The mass spectrum of copper is shown in **FIGURE 1.** **(CLO3, C4)**

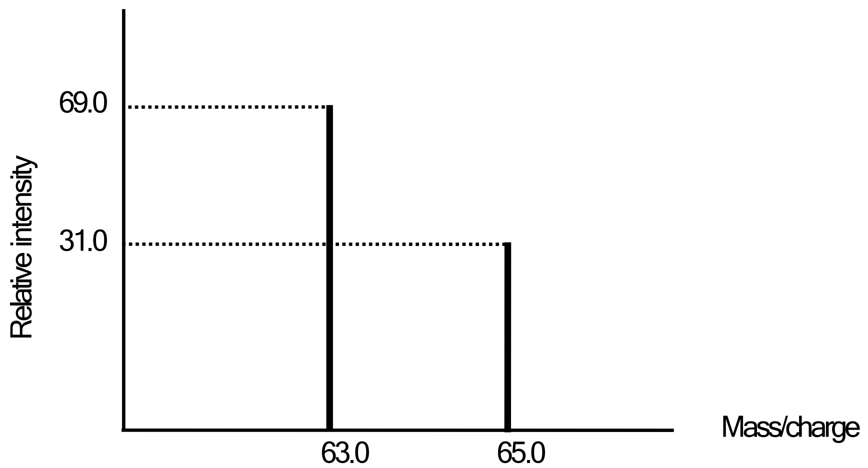

FIGURE 1

Based on **FIGURE 1**,
   a) Write all the isotopes of copper.
   b) Determine the percentage abundance of each isotope.
   c) Calculate the relative atomic mass of copper. *(63.62)*

5. **FIGURE 2** shows the mass spectrum of element ***X***:

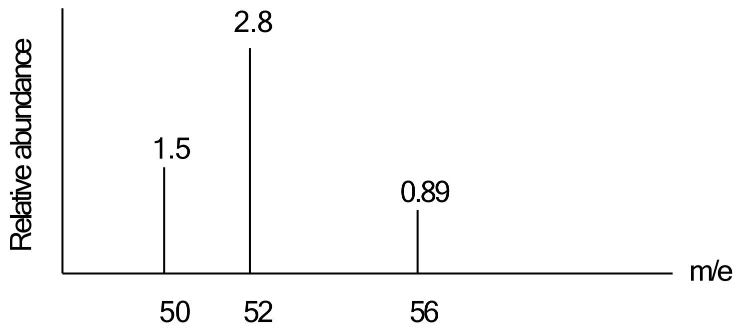

FIGURE 2

   a) State the number of protons and neutrons in isotope $^{52}_{27}X$
   b) Calculate the relative atomic mass of ***X***. *(52.11)* **MID SEM 2011 / 2012 (CLO3, C4)**

6. Element **J** has 10 protons. **FIGURE 3** shows the mass spectrum of **J**.

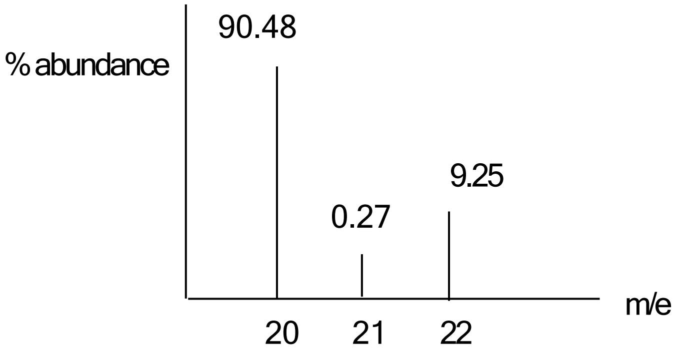

FIGURE 3

a) Write the isotopic notation of **J** with the highest percentage abundance.

b) Calculate the average atomic mass of **J**, if the isotopic masses are 19.9924 amu, 20.9938 amu and 21.9914 amu. *(20.18 amu)*

MID SEM DK015 2014 / 2015 (CLO3, C4)

HOUR: 2

7. Given the following table:

Isotope	Mass (amu)	Percentage abundance
^{107}Ag	106.91	51.50
^{109}Ag	108.90	48.50

Calculate the average atomic mass of silver. *(107.88 amu)*

MID SEM 2014 /2015 (CLO3, C4)

8. Iron consists of 5.82% ^{54}Fe, 91.66% ^{56}Fe, 2.19% ^{57}Fe and 0.33% ^{58}Fe. The isotopic masses of these four isotopes are 53.9396 u, 55.9394 u, 56.9354 u and 57.9333 u respectively. Calculate the relative atomic mass of iron. **(CLO3, C4)**

(55.85)

9. The two naturally occurring isotopes of gallium are gallium-69 and gallium-71 which are found in the ratio of 3:2. Determine the atomic mass of gallium. **MID SEM 2008 /2009 (CLO3, C4)**

(69.8 amu)

10. The isotopes of Ag occur naturally as ^{107}Ag and ^{109}Ag with their isotopic masses of 106.91 a.m.u and 108.87 a.m.u respectively. If the average atomic mass of Ag is 107.9 a.m.u, what would be the percentage abundance of these two isotopes? **(CLO3, C4)**

(abundance of ^{107}Ag =49.49%; abundance of ^{109}Ag =50.51%)

11. Magnesium is an element with a proton number of 12. In a sample of magnesium, it was found that magnesium atoms have three different nucleon numbers of 24, 25 and 26.

a) Write the symbol for one of these three atoms, showing its proton number and nucleon number.

b) A sample of magnesium contains 78.6% of atoms with a nucleon number of 24, 10.1% of atoms with a nucleon number of 25 and 11.3% of atoms with a nucleon number of 26. Calculate the relative atomic mass of magnesium. *(24.33)*

c) Sketch and label the expected mass spectrum of the magnesium.

PSPM 2013 / 2014 (CLO3, C4)

HOUR: 3

12. Analysis of a gaseous hydrocarbon compound gives the following percent composition by mass: 85.7% C and 14.3% H.
   a) Define empirical formula and molecular formula.
   b) Determine the empirical formula of the hydrocarbon.
   c) 0.25 g of this compound occupies a volume of 100 mL at STP. Determine the molar mass and the molecular formula of the hydrocarbon **(CLO3, C3)**
   (56.00 g mol^{-1})

13. Oxalic acid is often used to remove blood and rust stains. It contains 2.27% hydrogen, 26.65% carbon and 71.08% oxygen by mass. The molar mass of oxalic acid is 90.0 g mol^{-1}.
   a) Determine the empirical formula of oxalic acid.
   b) What is its molecular formula? **PSPM JAN 1999 (CLO3, C3)**

14. A complete combustion of a hydrocarbon forms 1.10 g of CO_2 and 0.45 g of H_2O. The molar mass is 84.00 g mol^{-1}. Determine the empirical formula and molecular formula of the hydrocarbon.
   (CLO3, C3)

15. A compound **P**, with the relative molecular mass of 46, contains the elements carbon, hydrogen and oxygen. When 2.30 g sample of **P** is completely burnt in excess oxygen, 4.40 g carbon dioxide and 2.70 g water were produced.
   a) Calculate the mass of carbon, hydrogen and oxygen in 2.30 g of **P**.
   b) Determine the empirical formula and molecular formula of **P**.
   c) Write a balanced equation for the combustion. **PSPM JAN 2000 (CLO3, C3)**
   (1.2 g C, 0.3 g H, 0.8 g O)

HOUR: 4

16. Define the terms: **(CLO1, C1)**
   a) molarity
   b) molality
   c) percentage by mass.

17. An aqueous solution contains 167 g $CuSO_4$ in 820 mL of solution. The density of the solution is 1.195 g mL^{-1}. Calculate the following:
   a) molarity of the solution *(1.28M)*
   b) percentage by mass of $CuSO_4$ *(17.04 %)*
   c) mole fraction of each component *(0.0226, 0.9774)*
   d) molality of the solution *(1.29 molal)*
   (CLO3, C4)

18. A certain amount of NaOH is dissolved in 98.0 g of water at 25 °C. The final volume of NaOH solution is 100 mL. Given the density of the solution is 1.33 g mL^{-1} at 25 °C, calculate the number of moles of NaOH used. *(0.875 mol)* **MID SEM 2010 / 2011 (CLO3, C4)**

19. A student would like to prepare 250 mL of 0.25 mol L^{-1} lead (II) nitrate, $Pb(NO_3)_2$ solution. Calculate the mass of $Pb(NO_3)_2$ required to prepare the solution. *(20.70 g)*
MID SEM 2015 / 2016 (CLO3, C4)

20. A 200 cm^3 of perfume contains 28 cm^3 of alcohol. What is the concentration of alcohol by volume in this solution? **(CLO3, C4)**
(14.00%)

21. Pure acetic acid, CH_3COOH, is a liquid with a density of 1.049 g mL^{-1}. A 10.0 mL of pure acetic acids is pipette into a 100 mL volumetric flask and distilled water is added to the mark. Calculate the molarity of the solution produced. **MID SEM 2013 / 2014 (CLO3, C4)**

[Molar mass of CH_3COOH = 60 g mol^{-1}] *(1.75 M)*

HOUR: 5

22. An aqueous ammonia solution with a density of 0.898 g mL^{-1} contains 28.0% ammonia by mass. Calculate the molarity of the solution. **MID SEM 2006 / 2007 (CLO3, C4)**
(14.79 M)

23. The density of an aqueous solution containing 10.0 % of ethanol, C_2H_5OH by mass is 0.984 g mL^{-1}. Calculate
   a) the molality of the solution *(2.42 molal)*
   b) its molarity *(2.14M)*
   c) the volume of the solution that contains 0.125 mole of ethanol *(0.058 L)*
   [molar mass of ethanol, C_2H_5OH = 46.0 g mol^{-1}] **MID SEM 2008 / 2009 (CLO3, C4)**

24. The density of 10.5 molal NaOH solution is 1.33 g mL^{-1} at 20.0 °C. Calculate: **(CLO3, C4)**
   a) the mole fraction of NaOH. *(0.159)*
   b) the percentage by mass of NaOH. *(29.58%)*
   c) the molarity of the solution. *(9.835 M)*

25. Sodium carbonate, Na_2CO_3 dissolves in ethanol to give 2.5 M solution. Calculate the molality of the solution if the density of the solution is 1.43 g mL^{-1}. **PSPM 2016 / 2017 (CLO3, C4)**
(2.15 mol kg^{-1})

26. A 50.0 mL saturated solution of NaOH containing 52.0% NaOH by weight with a density of 1.48 g mL^{-1} is used to prepare a 0.100 M solution. Determine the initial concentration of the NaOH solution and the amount of water required to prepare the 0.100 M NaOH solution.
PSPM 2006 / 2007 (CLO3, C4)
(19.24 M, 9.57 L @ 9570 mL)

HOUR: 6

27. Balance each of the following equations: **(CLO3, C3)**
   a) $NaOH(aq) + FeCl_3(s) \longrightarrow Fe(OH)_3(s) + NaCl(aq)$
   b) $C_4H_{10}(g) + O_2(g) \longrightarrow CO_2(g) + H_2O(l)$
   c) $Fe(s) + H_2O(l) \longrightarrow Fe_3O_4(s) + H_2(g)$
   d) $Fe_2O_3(s) + HCl(aq) \longrightarrow FeCl_3(aq) + H_2O(l)$
   e) $Cr(OH)_3(aq) + IO_3^-(aq) \longrightarrow CrO_3^{2-}(aq) + I^-(aq)$ (acidic medium)
   f) $Cl_2(aq) \longrightarrow ClO_4^-(aq) + Cl^-(aq)$ (basic medium)

28. A solution containing I^- ion was titrated in acidic medium with MnO_4^- ion to form I_2 and Mn^{2+}.
    a) Write the balanced equation for the titration.
    b) How many millilitres of a 0.206 M I^- solution are needed to reduce 22.5 mL of a 0.374 M MnO_4^-? *(204.25 mL)*

(CLO3, C4)

HOUR: 7

29. A reaction between 7.0 g of copper(II) oxide and 50 mL of 0.20 M nitric acid produces copper(II) nitrate, $Cu(NO_3)_2$ and water. **(CLO3, C4)**
    a) Define limiting reactant.
    b) Write the balanced chemical equation for the above reaction
    c) Determine the limiting reactant
    d) Calculate the mass of excess reactant after the reaction *(6.60g)*

30. At room temperature, 255 g of copper (I) oxide, Cu_2O react with 180 g of copper (I) sulphide Cu_2S according to the equation below,

$$2Cu_2O \; + \; Cu_2S \longrightarrow 6Cu \; + \; SO_2$$

Determine:
    a) the limiting reactant.
    b) the mass of the excess reactant remained at the end of the reaction. *(38.07g)*
    (molar mass of Cu_2O = 143.2 g mol^{-1} ; molar mass of Cu_2S = 159.3 g mol^{-1})

MID SEM 2006 / 2007 (CLO3, C4)

31. In an experiment, 0.355 g of aluminium, Al reacts with 20.00 mL of 4.00 M hydrochloric acid, HCl according to the following equation:

$$2Al\,(s) + 6HCl\,(aq) \longrightarrow 2AlCl_3\,(aq) + 3H_2\,(g)$$

    a) Determine the limiting reactant
    b) If the percentage yield of hydrogen is 85.0%, calculate the actual mass of hydrogen obtained. *(0.0335 g)*

MID SEM 2010 / 2011 (CLO3, C4)

32. In an experiment, 1.46g of magnesium is added into 160.00 mL of 0.50 mol L^{-1} hydrochloric acid. The reaction involved is:

$$Mg\,(s) + 2HCl\,(aq) \rightarrow MgCl_2\,(aq) + H_2\,(g)$$

    a) Determine the limiting reactant for the above reaction
    b) Calculate the percentage yield if 672.00 mL of hydrogen gas is obtained at STP. *(75.00 %)*

MID SEM 2014 / 2015 (CLO3, C4)

HOUR: 8

33. What is meant by percentage yield?
The combustion of potassium chloride, KCl produces potassium chlorate, $KClO_3$ according to the equation

$$2KCl\,(s) \; + \; 3O_2\,(g) \longrightarrow 2KClO_3\,(s)$$

Calculate the mass of oxygen gas needed to produce 3.80 g of potassium chlorate if the percentage yield is 86.0%

PSPM JUNE 2000 (CLO3, C4)
(1.73 g)

34. A 72.0 g vanadium pentoxide, V_2O_5, reacts with excess aluminium, Al, at high temperature to produce vanadium metal, V, and aluminium oxide, Al_2O_3. [Ar V : 51]
   a) Write a chemical equation for the above reaction.
   b) Calculate the mass of vanadium produced. *(40.39 g)*

MID SEM 2013 / 2014 (CLO3, C4)

35. Adipic acid, $H_2C_6H_8O_4$, is produced by a reaction between cyclohexane and excess oxygen. The equation for the reaction is:

$$2C_6H_{12}\,(l) \;+\; 5O_2\,(g) \longrightarrow 2H_2C_6H_8O_4\,(l) \;+\; 2H_2O\,(l)$$

   a) If 45.0 g of cyclohexane is used, calculate the theoretical yield of adipic acid *(78.21 g)*
   b) Given the actual yield of adipic acid is 63.5 g, what is the percentage yield of adipic acid?

(81.19 %) **MID SEM 2012 /2013 (CLO3, C4)**

36. In an experiment, a student allowed benzene C_6H_6 to react with excess bromine, Br_2 in an attempt to prepare bromobenzene, C_6H_5Br. On the basis of the equation: **(CLO3, C4)**

$$C_6H_6 \;+\; Br_2 \longrightarrow C_6H_5Br \;+\; HBr$$

   a) What is the maximum number of grams of C_6H_5Br the student could have hoped to obtain from 15.00g of benzene? *(30.17 g)*
   b) Calculate the percentage yield if 28.5g of C_6H_5Br is obtained. *(94.46 %)*
   c) With the same percentage yield, how many grams of benzene C_6H_6 will have to start with to obtain 10.0 g of C_6H_5Br. *(5.27 g C_6H_6)*

37. The neutralisation reaction between sulphuric acid and sodium hydroxide yields sodium sulphate and water.
   a) Write a balanced equation for the neutralisation.
   b) Calculate the moles of sodium hydroxide needed to yield 0.080 mol of sodium sulphate. *(0.16 mol)*
   c) Calculate the mass of sulphuric acid needed for the reaction in (b). *(7.85 g)*

PSPM JUNE 2000 (CLO3, C4)